

Some Properties of Cyclotomic Polynomials

Mobius Inversion Formula

Define the *Mobius Function* $\mu: \mathbb{Z}^+ \rightarrow \{0, 1, -1\}$ as follows:

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^k & n \text{ is square free and } k \text{ is the number of prime divisor of } n. \\ 0 & \text{otherwise} \end{cases}$$

Clearly μ is *multiplicative*, namely if m and n are relatively prime, then $\mu(mn) = \mu(m)\mu(n)$. Moreover

Theorem 1: Let n be a positive integer. Then $\sum_{d|n} \mu(d) = 1$ for $n = 1$, and

$$\sum_{d|n} \mu(d) = 0 \text{ for } n \geq 2.$$

Proof. For $n = 1$, it is clear. Now define $T = \prod_{\substack{p|n \\ p \text{ prime}}} p$ as the product of all distinct prime divisors of n . Then it is clear $\sum_{d|n} \mu(d) = \sum_{d|T} \mu(d)$. But suppose now T has k distinct prime divisors, i.e., $T = p_1 p_2 \dots p_k$, then

$$\sum_{d|T} \mu(d) = \sum_{l=0}^k \binom{k}{l} (-1)^l = (1-1)^k = 0.$$

Now we have

Theorem 2 (Mobius Inversion Formula): Suppose $F, f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ are functions so that $F(n) = \sum_{d|n} f(d)$. Then $f(n) = \sum_{d|n} \mu(d) F(\frac{n}{d})$.

Proof. Note $\sum_{d|n} \mu(d) F(\frac{n}{d}) = \sum_{d|n} \mu(d) \sum_{\substack{t|n \\ t|\frac{n}{d}}} f(t)$. Now t is a divisor of $\frac{n}{d}$ and is a

divisor of n . Thus

$$\sum_{d|n} \mu(d) \sum_{\substack{t|\frac{n}{d}}} f(t) = \sum_{t|n} f(t) \sum_{\substack{d|n \\ t|\frac{n}{d}}} \mu(d) = \sum_{t|n} f(t) \sum_{\substack{d|\frac{n}{t}}} \mu(d). \text{ But the term } \sum_{\substack{d|\frac{n}{t}}} \mu(d) \text{ is } 1 \text{ when}$$

$n = t$ and 0 otherwise. Hence we are done.

Remark: The multiplicative version of the theorem is as follows. Given f ,

$F: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ such that $F(n) = \prod_{d|n} f(d)$, then $f(n) = \prod_{d|n} F(\frac{n}{d})^{\mu(d)}$.

Proof. $\prod_{d|n} F(\frac{n}{d})^{\mu(d)} = \prod_{d|n} (\prod_{t|\frac{n}{d}} f(t))^{\mu(d)} = \prod_{t|n} f(t)^{\sum_{d|\frac{n}{t}} \mu(d)} = \prod_{t|n} f(t)^{\sum_{d|\frac{n}{t}} \mu(d)} = f(n)$.

Primitive Roots of Unity

A complex number ξ is an n th root of unity if $\xi^n = 1$. We know they are $e^{\frac{2\pi i}{n}}, e^{\frac{4\pi i}{n}}, \dots, e^{\frac{2n\pi i}{n}}$, where $e^{\frac{2k\pi i}{n}} = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}$. If ξ is an n th root of unity, then the smallest positive integer k such that $\xi^k = 1$ is the *order* of ξ , denoted by $\text{ord}(\xi)$. Easy to show the **Facts**:

1. If ξ is an n th root of unity and $\xi^k = 1$, using division algorithm, $\text{ord}(\xi) | k$. In

particular $\text{ord}(\xi) | n$.

2. Let ξ be an n th root of unity, then $\xi^k = \xi^l$ if and only if $k \equiv l \pmod{\text{ord}(\xi)}$. This simply follows from $\xi^{k-l} = 1$.

Now we define: if ξ is an n th root of unity. Then ξ is a **primitive n th root of unity** if $\text{ord}(\xi) = n$. Indeed if $\text{ord}(\xi) = d$, then ξ is a primitive d root of unity. We now observe other **Facts**:

3. If ξ is a primitive n th root of unity, then $\xi, \xi^2, \dots, \xi^n = 1$ are all n th (distinct) roots of unity. They are all distinct roots of $x^n = 1$.

4. If ξ is a primitive n th root of unity, then ξ^k is a primitive n th root of unity if and only if $\gcd(n, k) = 1$. Hence there are $\phi(n)$ (Euler's totient function) of them.

Indeed suppose $\text{ord}(\xi^k) = d$, then $\xi^{kd} = 1$, hence $n | kd$. So if $\gcd(n, k) = 1$, then

$n | d$, thus $n = d$. On the other hand, $\gcd(n, k) \neq 1$, then $\xi^{\frac{k}{\gcd(n, k)} \frac{n}{\gcd(n, k)}} = 1$, so $d < n$, or ξ^k not primitive.

5. Again assume ξ is a primitive n th root of unity, t a factor of n , then ξ^t is of order $\frac{n}{t}$, and there are $\phi(\frac{n}{t})$ elements of such order. This gives a proof of

$$\sum_{d|n} \phi(d) = n.$$

Example 1: From $\sum_{d|n} \phi(d) = n$, and the Inversion formula, we have

$$\phi(n) = \sum_{d|n} \mu\left(\frac{n}{d}\right) d = \sum_{t|n} \mu(t) \frac{n}{t} = n \sum_{t|n} \frac{\mu(t)}{t}.$$
 Suppose now $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r}$, then

$$\begin{aligned} \sum_{t|n} \frac{\mu(t)}{t} &= 1 - \sum_i \frac{1}{p_i} + \sum_{i < j} \frac{1}{p_i p_j} - \dots + (-1)^r \frac{1}{p_1 p_2 \dots p_r} \\ &= \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_r}\right). \end{aligned}$$

We then have a formula of the Euler's totient function.

Cyclotomic Polynomials

The ***n*th cyclotomic polynomial**, $\Phi_n(x)$, is the (monic) polynomial having exactly the primitive *n*th roots of unity as roots, that is,

$$\Phi_n(x) = \prod_{\substack{\xi^n = 1 \\ \text{ord}(\xi) = n}} (x - \xi).$$

As discussed, we see

Theorem 3: $x^n - 1 = \prod_{d|n} \Phi_d(x).$

Check that the two sides have the same roots and both are monic. Comparing the x^n terms, we also have $n = \sum_{d|n} \phi(d).$

Using the Inversion Formula, we have

Theorem 4: $\Phi_n(x) = \prod_{d|n} (x^{\frac{n}{d}} - 1)^{\mu(d)}.$

Example 2: Using Theorem 3 and 4, we get

$$\Phi_1(x) = x - 1$$

$$\Phi_2(x) = \frac{x^2 - 1}{x - 1} = x + 1$$

$$\Phi_3(x) = \frac{x^3 - 1}{x - 1} = x^2 + x + 1$$

$$\Phi_4(x) = \frac{(x^4 - 1)(x - 1)^0}{x^2 - 1} = x^2 + 1$$

$$\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$$

$$\Phi_6(x) = x^2 - x + 1$$

$$\Phi_7(x) = x^6 + x^5 + \dots + 1$$

$$\Phi_8(x) = x^4 + 1$$

$$\Phi_9(x) = x^6 + x^3 + 1$$

$$\Phi_{10}(x) = x^4 - x^3 + x^2 - x + 1$$

$$\Phi_{11}(x) = x^{10} + x^9 + \dots + x + 1$$

$$\Phi_{12}(x) = x^4 - x^2 + 1$$

$$\Phi_{13}(x) = x^{12} + x^{10} + \dots + x + 1$$

$$\Phi_{14}(x) = x^6 - x^5 + x^4 - x^3 + x^2 - x + 1$$

$$\Phi_{15}(x) = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1.$$

Example 3:

1. For an odd prime p , $\Phi_p(x) = x^{p-1} + x^{p-2} + \dots + 1$.

2. For prime power p^r , $\Phi_{p^r}(x) = \frac{x^{p^r} - 1}{x^{p^{r-1}} - 1} = x^{p^{r-1}(p-1)} + x^{p^{r-1}(p-2)} + \dots + x^{p^{r-1}} + 1$.

3. If n is odd larger than 1, then $\Phi_{2n}(x) = \Phi_n(-x)$. In particular for an odd prime p ,

$$\Phi_{2p}(x) = x^{p-1} - x^{p-2} + \dots - x + 1.$$

We give some properties of cyclotomic polynomials. Before all that, a lemma:

Lemma 1: Suppose $f(x) = x^m + a_{m-1}x^{m-1} + \dots + a_0$ and

$g(x) = x^n + b_{n-1}x^{n-1} + \dots + b_0$ are polynomials with rational coefficients. If all coefficients of $f(x)g(x)$ are integers, then so are $f(x)$ and $g(x)$.

Proof. Let M and N be the smallest positive integers so that all coefficients of $Mf(x)$ and $Ng(x)$ are integers. Let $A_i = Ma_i, 0 \leq i \leq m-1$ and $A_m = M$; also $B_j = Nb_j, 1 \leq j \leq n-1, B_n = N$. Then $MNf(x)g(x) = A_m B_n x^{m+n} + \dots + A_0 B_0$ will consist only integer coefficients. Suppose $MN > 1$ and p a prime divisor of MN . We claim there exists an $i, 0 \leq i \leq m$, such that $p \nmid A_i$. Indeed if $p \nmid M$, then $p \nmid A_m$ and if $p \mid M$, then $p \mid A_i$ for all $i, 0 \leq i \leq m$, then $\frac{A_i}{p} = \frac{M}{p} a_i$ an integer, contradicting the minimality of M . Likewise there exists an integer j such that $p \nmid B_j$. Let I and J be the greatest of these i and j . Then the coefficient of x^{I+J} in $mnf(x)g(x)$ is $A_I B_J + pR$ not divisible by p , contradiction again.

Theorem 5:

1. The coefficients of $\Phi_n(x)$ are integers.
2. $\Phi_n(0) = 1$.
3. $\Phi_n(x) > 0$ for all real x .
4. $\Phi_n(x)$ is strictly increasing for $x > 1$.
5. For all $x > 0$, $(x-1)^{\phi(n)} < \Phi_n(x) < (x+1)^{\phi(n)}$.
6. Let $n \geq 2$. Let $\Phi_n(x) = \sum_{i=0}^m c_i x^i$. Then (i) $\Phi_n(x) = x^m \Phi_n(1/x)$ for $x \neq 0$, (ii) $c_{m-r} = c_r$ for $0 \leq r \leq m$. (Here of course $m = \phi(n)$.)
7. $\Phi_n(x)$ are irreducible over $\mathbb{Z}$.

Proof.

1. By induction $\Phi_1(x) = x-1$. Suppose that the statement is true for all $k < n$. Then

$$\Phi_n(x) = \frac{x^n - 1}{\prod_{\substack{d|n, \\ d \neq n}} \Phi_d(x)}.$$

The coefficients of $\Phi_n(x)$ are rational and thus are integers

by Lemma 1.

2. We note the roots of $x^n - 1$ are 1, -1 (when n even) and complex roots appear in pairs $e^{\frac{2k\pi i}{n}} = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}$ and $e^{\frac{2(n-k)\pi i}{n}} = \cos \frac{2(n-k)\pi}{n} + i \sin \frac{2(n-k)\pi}{n}$.

(Note they are of the same order),

hence $(x - e^{\frac{2k\pi i}{n}})(x - e^{\frac{2(n-k)\pi i}{n}}) = x^2 - 2x \cos \frac{2k\pi}{n} + 1$. So $\Phi_n(0) = 1$.

3. 4. For $-1 < a < 1$, write $f_a(x) = x^2 - 2ax + 1$. Then

$x^2 - 2ax + 1 = (x - a)^2 + (1 - a^2) > 0$ and is strictly increasing for all $x \geq 1$. Thus (3) and (4).

5. Also $(x - 1)^2 < x^2 - 2ax + 1 < (x + 1)^2$ for all $x > 0$. Thus (5).

6. The statement is true for $n = 2$. For $n > 2$, write $a_k = \cos \frac{2k\pi}{n}$, and by putting

in suitable pairs, we have

$x^m \Phi_n\left(\frac{1}{x}\right) = \prod x^2 \left(1 - \frac{2a_k}{x} + \frac{1}{x^2}\right) = \prod (x^2 - 2a_k x + 1) = \Phi_n(x)$. Hence

$\sum_{i=0}^m c_i x^i = \sum_{i=0}^m c_i x^{m-i} = \sum_{j=0}^m c_{m-j} x^j = \sum_{i=0}^m c_{m-i} x^i$ for all $x \neq 0$. Thus $c_r = c_{m-r}$

(palindrome).

7. See Jacobson or Jameson. For $n = p$, may use Eisenstein. Thus we say $\Phi_n(x)$ is the *minimal polynomial* of a primitive n th root of unity ξ .

Theorem 6.

1. Let p be a prime number and n a positive integer. Then

$$\Phi_{pn}(x) = \begin{cases} \Phi_n(x^p) & \text{if } p \mid n \\ \frac{\Phi_n(x^p)}{\Phi_n(x)} & \text{if } p \nmid n. \end{cases}$$

2. Let p be a prime number, n and k are positive integer. Then

$$\Phi_{p^k n}(x) = \begin{cases} \Phi_n(x^{p^k}) & \text{if } p \mid n \\ \frac{\Phi_n(x^{p^k})}{\Phi_n(x^{p^{k-1}})} & \text{if } p \nmid n. \end{cases}$$

Proof. 1. Suppose $p \mid n$. Then

$$\begin{aligned}
\Phi_{pn}(x) &= \prod_{d|pn} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \\
&= \left(\prod_{d|n} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \right) \left(\prod_{\substack{d|pn \\ d \nmid n}} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \right) \\
&= \Phi_n(x^p) \prod_{\substack{d|pn \\ d \nmid n}} (x^{\frac{pn}{d}} - 1)^{\mu(d)}.
\end{aligned}$$

As $d|pn$, $d \nmid n$ and $p|n$, implies that $p^2|d$, so $\mu(d) = 0$. Hence the result.

Suppose now $p \nmid n$. Then

$$\begin{aligned}
\Phi_{pn}(x) &= \prod_{d|pn} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \\
&= \left(\prod_{d|n} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \right) \left(\prod_{d|p} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \right) \\
&= \left(\prod_{d|n} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \right) \left(\prod_{d|n} (x^{\frac{n}{d}} - 1)^{-\mu(d)} \right) \\
&= \frac{\Phi_n(x^p)}{\Phi_n(x)}.
\end{aligned}$$

2. Hence

$$\Phi_{p^k n}(x) = \Phi_{p^{k-1}n}(x^p) = \dots = \Phi_{pn}(x^{p^{k-1}}) = \begin{cases} \Phi_n(x^{p^k}) & \text{if } p|n \\ \frac{\Phi_n(x^{p^k})}{\Phi_n(x^{p^{k-1}})} & \text{if } p \nmid n. \end{cases}$$

Lemma 2:

1. Let p be a prime number. Suppose $x^n - 1$ has a double root modulo p , i.e., $x^n - 1 \equiv (x-a)^2 f(x) \pmod{p}$. Then $p|n$. (Here a is an integer and $f(x)$ is a polynomial with integer coefficients.)

2. Let $d|n$ and $d < n$, x be an integer. Suppose p is a common divisor of $\Phi_d(x)$

and $\Phi_n(x)$. Then $p|n$.

Proof. 1. Clearly $p \nmid a$, otherwise $-1 \equiv 0 \pmod{p}$. Taking derivatives, get

$$nx^{n-1} \equiv 2(x-a)f(x) + (x-a)^2 f'(x) \pmod{p}, \text{ or } na^{n-1} \equiv 0 \pmod{p}. \text{ Hence } p|n.$$

2. $x^n - 1 = \prod_{d|n} \Phi_d(x)$, and p divides $\Phi_d(x)\Phi_n(x)$, thus $x^n - 1$ has a double root and the result.

Theorem 7

1. Let x be an integer. Then every prime factor p of $\Phi_n(x)$ either satisfies $p \equiv 1 \pmod{n}$ or $p|n$.
2. Let p be a prime number. Then every prime factor q of $1 + x + \dots + x^{p-1}$ either satisfies $q \equiv 1 \pmod{p}$ or $q = p$.

Proof.

1. Let p be a prime divisor of $\Phi_n(x)$. Note $p \nmid x$ since $p|\Phi_n(x)|x^n - 1$. Let $k = \text{ord}_n(x)$. Since $p|x^n - 1$, we have $x^n \equiv 1 \pmod{p}$, so $k|n$. If $k = n$, then $n|p-1$, i.e., $p \equiv 1 \pmod{n}$, since $x^{p-1} \equiv 1 \pmod{p}$ by Fermat' Little Theorem. Suppose now $k < n$. Since $0 \equiv x^k - 1 = \prod_{d|k} \Phi_d(x) \pmod{p}$, there exists a divisor d of k such that $p|\Phi_d(x)$. But $d|k|n$ and $d < n$, so from Lemma 2.2, the result follows.

2. Note $\Phi_p(x) = 1 + x + x^2 + \dots + x^{p-1}$.

Example 4 Check the prime factors of $1 + x + \dots + x^4 = \Phi_5(x)$ are divisible by 5 or

$\equiv 1 \pmod{5}$. Also $\Phi_8(x) = x^4 + 1$. The prime factors are either divisible by 2 or

$\equiv 1 \pmod{8}$. Check $8^4 + 1 = 17 \times 241$. Now $\Phi_{72}(x) = x^{24} - x^{12} + 1$. Check

$2^{24} - 2^{12} + 1 = 433 \times 38737$ and each prime factor $\equiv 1 \pmod{72}$.

Theorem 8

1. Let a and b be positive integers and x an integer. Then $\gcd(x^a - 1, x^b - 1) = |x^{\gcd(a,b)} - 1|$.
2. Let a and b be positive integers. Suppose $\gcd(\Phi_a(x), \Phi_b(x)) > 1$ for some

integer x . Then $\frac{a}{b} = p^k$, the integral power of a prime number.

Proof.

1. Let $T = \gcd(x^a - 1, x^b - 1)$ and $t = \gcd(a, b)$. Since $x^t - 1$ divides both $x^a - 1$ and $x^b - 1$. Thus $x^t - 1 \mid T$. Now as $\gcd(x, T) = 1$. Let $d = \text{ord}_T(x)$. Then $T \mid x^d - 1$ and since $x^a \equiv x^b \equiv 1 \pmod{T}$, we have $d \mid a$ and $d \mid b$, so $d \mid t$. Thus $x^d - 1 \mid x^t - 1$, so $T \mid x^t - 1$. Together we have the result.

2. There are several steps to prove the statement. Suppose p is a common prime divisor of $\Phi_a(x)$ and $\Phi_b(x)$, we want to show $\frac{a}{b}$ is a power of p .

Now let $a = p^\alpha A$ and $b = p^\beta B$, where $\alpha, \beta \geq 0$ and A and B are not divisible by p , we want to show $A = B$.

Clearly $p \mid \Phi_a(x) \mid x^a - 1$, thus $p \nmid x$.

(Step 1) We now want to show in fact $p \mid \Phi_A(x)$. This is trivial if $\alpha = 0$.

Otherwise $\alpha > 0$. By Theorem 6.2, we have

$$0 \equiv \Phi_a(x) = \frac{\Phi_A(x^{p^\alpha})}{\Phi_A(x^{p^{\alpha-1}})} \pmod{p},$$

So $\Phi_A(x^{p^\alpha}) \equiv 0 \pmod{p}$. But $x^{p^\alpha} \equiv x \cdot x^{p^{\alpha-1}} = x \cdot x^{(p-1)s} \equiv x \pmod{p}$, where s is

some integer. Thus $\Phi_A(x) \equiv 0 \pmod{p}$. Likewise $p \mid \Phi_B(x)$.

(Step 2) Suppose $A > B$. Let $t = \gcd(A, B)$. Then $t < A$. Now $p \mid \Phi_A(x) \mid x^A - 1$

and $p \mid \Phi_B(x) \mid x^B - 1$, so $p \mid \gcd(x^A - 1, x^B - 1) = x^t - 1$. But

$0 \equiv x^t - 1 = \prod_{d \mid t} \Phi_d(x) \pmod{p}$, thus there exists a divisor d of t such that $p \mid \Phi_d(x)$.

But $d \mid t \mid A$, $d < A$ and $p \mid \Phi_A(x)$, so by Theorem 7.1, $p \mid A$, a contradiction.

Likewise $A < B$ is not true. Thus $A = B$.

Applications

Theorem 9: Let d be a positive integer, there exist infinitely many primes p in the arithmetical progression $1, 1+d, 1+2d, \dots, 1+kd, \dots$.

Proof. Suppose not, let $p_1, p_2, \dots, p_r$ be all such primes such that $p_i \equiv 1 \pmod{d}$ for $i = 1, 2, \dots, r$. Define $N = dp_1 p_2 \dots p_r$ so $|\Phi_d(N)| > 1$. Now let p be a prime such that $p \nmid \Phi_d(N)$. Since $\Phi_d(N) \equiv \Phi_d(0) \equiv 1 \pmod{N}$, $p \nmid N$, $p \nmid d$. Thus by Theorem 7.1, $p \equiv 1 \pmod{d}$, but clearly p is not one of the p_i 's.

Example 5 (IMO2002 Shortlist). Let $p_1, p_2, \dots, p_n$ be distinct primes greater than 3. Then $2^{p_1 p_2 \dots p_n} + 1$ has at least 2^{n-1} divisors.

Proof. It suffices to show that $2^{p_1 p_2 \dots p_n} + 1$ has at least 2^{n-1} pairwise relatively prime divisors and hence at least 2^{n-1} divisors. We know

$$\begin{aligned} (2^{p_1 p_2 \dots p_n} + 1)(2^{p_1 p_2 \dots p_n} - 1) &= 2^{2 p_1 p_2 \dots p_n} - 1 = \prod_{d|2 p_1 p_2 \dots p_n} \Phi_d(2) \\ &= \left(\prod_{d|p_1 p_2 \dots p_n} \Phi_d(2) \right) \left(\prod_{d|p_1 p_2 \dots p_n} \Phi_{2d}(2) \right) \\ &= (2^{p_1 p_2 \dots p_n} - 1) \left(\prod_{d|p_1 p_2 \dots p_n} \Phi_{2d}(2) \right), \end{aligned}$$

and hence $2^{p_1 p_2 \dots p_n} + 1 = \prod_{d|p_1 p_2 \dots p_n} \Phi_{2d}(2)$.

From Theorem 8.2, if $\Phi_a(2)$ and $\Phi_b(2)$ are not relatively prime, then a/b is a prime power. We just have to find 2^{n-1} divisors of $p_1 p_2 \dots p_n$ so that no two of them differ by exactly one prime number. We just to take those divisors who contain even number of prime divisors, and there are 2^{n-1} of them.

Example 6 (IMO2006 Shortlist). Find all integer solutions of the equation

$$\frac{x^7 - 1}{x - 1} = y^5 - 1.$$

The equation is equivalent to

$$1 + x + x^2 + \dots + x^6 = (y - 1)(1 + y + \dots + y^4).$$

From Theorem 7.2, if p divides $1 + x + \dots + x^6$, then either $p = 7$ or $p \equiv 1 \pmod{7}$. Thus $y - 1 \equiv 0 \pmod{7}$ or $y - 1 \equiv 1 \pmod{7}$; hence $y \equiv 1 \pmod{7}$ or $y \equiv 2 \pmod{7}$. If

$y \equiv 1 \pmod{7}$, then $1 + y + \dots + y^4 \equiv 5 \not\equiv 0, 1 \pmod{7}$, a contradiction. If $y \equiv 2 \pmod{7}$, then $1 + y + \dots + y^4 \equiv 31 \equiv 3 \not\equiv 0, 1 \pmod{7}$, another contradiction. Thus the equation has no solution.

Example 7 (USAMO1977). Determine all pairs of integers (m, n) such that

$1 + x^n + x^{2n} + \dots + x^{mn}$ is divisible by $1 + x + x^2 + \dots + x^m$.

The expressions are equivalent to $\frac{x^{n(m+1)} - 1}{x^n - 1}$ and $\frac{x^{m+1} - 1}{x - 1}$, respectively. Thus

$$\frac{(x^{n(m+1)} - 1)(x - 1)}{(x^{m+1} - 1)(x^n - 1)}$$

a polynomial. Now the roots of $x^{n(m+1)} - 1$ are distinct, so $x^{m+1} - 1$ and $x^n - 1$ have no common factor other than $(x - 1)$, hence $\gcd(m+1, n) = 1$. The condition is sufficient since $x^{n(m+1)} - 1 = (x^n)^{m+1} - 1 = (x^{m+1})^n - 1$ thus divisible by both $x^n - 1$ and $x^{m+1} - 1$.

Example 8 (IMO1995.6) Let p be an odd prime number. Find the number of subsets

A of the set $\{1, 2, \dots, 2p\}$ such that

- (a) A has exactly p elements, and
- (b) the sum of all the elements in A is divisible by p .

Let $\xi = e^{2\pi i/p}$ be a primitive root of unity. The idea is to consider products of p elements within the list $\xi^1, \xi^2, \dots, \xi^{2p}$.

Then

$$\prod_{i=1}^{2p} (x - \xi^i) = (x^p - 1)^2 = x^{2p} - 2x^p + 1.$$

By Viète's formula applied to the coefficient of x^p , get

$$2 = \sum_{1 \leq i_1 \leq \dots \leq i_p \leq 2p} \xi^{i_1 + \dots + i_p} = \sum_{j=0}^{p-1} n_j \xi^j,$$

Where n_j is the number of subsets $\{i_1, i_2, \dots, i_p\}$ of $\{1, 2, \dots, 2p\}$ such that

$$i_1 + i_2 + \dots + i_p = j \pmod{p}.$$

We look for the number n_0 .

But ξ is also a root of the polynomial of degree $p-1$, namely

$$P(x) = n_0 - 2 + \sum_{j=1}^{p-1} n_j x^j.$$

On the other hand, the minimal polynomial of ξ is the cyclotomic polynomial

$$\phi_p(x) = 1 + x + x^2 + \dots + x^{p-1}.$$

It follows that they differ by a constant c ,

$$n_0 - 2 + \sum_{j=1}^{p-1} n_j x^j = c(1 + x + x^2 + \dots + x^{p-1}).$$

By setting $x = 1$, we find

$$c = \frac{1}{p} \left(\sum_{j=0}^{p-1} n_j - 2 \right) = \frac{1}{p} \left(\binom{2p}{p} - 2 \right).$$

Using c , we obtain the value of n_0 by comparing the constant terms:

$$n_0 = 2 + \frac{1}{p} \left(\binom{2p}{p} - 2 \right).$$

Exercises

1. p a prime and m a positive integer. If $p \nmid m$, then $\Phi_{pm}(x)\Phi_m(x) = \Phi_m(x^p)$.
2. n odd, ω a primitive n th root of unity if and only if $-\omega$ a primitive $2n$ th root of unity. Hence $\Phi_{2n}(x) = \Phi_n(-x)$.
3. p a prime and k a positive integer, then $\Phi_{p^k}(1) = p$. (Use Theorem 6.1).
4. (Wedderburn) Every finite division ring (every nonzero element has an inverse) is a field (commutative).

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